Technical Supplement

SABER: Support-Conditioned Adaptive Bellman Error Reweighting for Sparse-Reward Hindsight Experience Replay

Anonymous Submission

Overview. This supplement gives a complete mathematical specification and pseudocode for SABER, followed by experimental settings, seed-level primary results, sensitivity and scope controls, support-signal diagnostics, runtime measurements, and an index of the anonymous code-and-data artifacts.

Working-draft completion status. One result slot remains *pending*: FPP DDPG Actor+Critic. No value is inferred from its incomplete run. This marker and paragraph must be removed after a complete, validated five-seed artifact replaces the remaining slot.

1 Complete SABER Specification

1.1 Scope, Problem Setting, and Notation

SABER is a critic-loss allocation rule for replay-based off-policy actor–critic learning. It is neither a replay sampler nor a network module. Consider a finite-horizon goal-conditioned MDP

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{G}, P, r, \gamma, H).$$

The agent receives observation o_t, desired goal g, and network input $x_t = (o_t, g)$. The achieved-goal map is $m : \mathcal{S} \to \mathcal{G}$. The sparse reward used in the experiments is

$$r(s_t, a_t, s_{t+1}, g) = \begin{cases} 0, & d(m(s_{t+1}), g) \le \epsilon_g, \\ -1, & \text{otherwise.} \end{cases} \tag{1}$$

Replay stores the executed action a_i^r. We reserve a_i^π for the detached current-policy action used only by SABER's discrete support query. There are $n_Q = 1$ critics for DDPG and $n_Q = 2$ for TD3 and SAC.

1.2 Future-Goal HER Batch and Unchanged Bellman Targets

Let an episode contain states $s_0, \dots, s_T$ and transitions indexed by $t = 0, \dots, T-1$. When transition t is selected for relabeling and has an eligible later stored transition, the implemented future-goal convention samples

$$\begin{aligned} u &\sim \mathrm{Unif}\{t+1, \dots, T-1\}, \\ \widetilde{g}_i &= m(s_{u+1}), \\ \widetilde{r}_i &= r(s_t, a_t, s_{t+1}, \widetilde{g}_i). \end{aligned} \tag{2}$$

If the eligible set is empty, or if relabeling is not selected, the original desired goal and its corresponding reward are retained. Thus the nominal relabeling probability is $p_{\mathrm{HER}} = 0.8$, while the realized fraction can be slightly lower near an episode boundary. The relabeled network inputs are

$$x_i = (o_i, \widetilde{g}_i), \qquad x_i' = (o_i', \widetilde{g}_i).$$

Let $\chi_i \in \{0, 1\}$ be the bootstrap mask. Successful relabeled transitions and time-limit transitions use $\chi_i = 1$; genuine terminating transitions use $\chi_i = 0$. For backbone A, define

$$y_i^{\mathsf{A}} = \widetilde{r}_i + \gamma \chi_i V_i^{\mathsf{A}}(x_i'), \tag{3}$$

where target values are evaluated without gradient. Writing barred parameters for target networks, the implemented continuation values have the standard forms

$$V_i^{\mathrm{DDPG}} = Q_{\bar{\phi}_1}(x_i', \mu_{\bar{\theta}}(x_i')), \tag{4}$$

$$\begin{aligned} \varepsilon_i &\sim \mathrm{clip}\big(\mathcal{N}(0, \sigma_{\mathrm{targ}}^2 I), -c, c\big), \\ \widetilde{a}_i' &= \mathrm{clip}_{\mathcal{A}}(\mu_{\bar{\theta}}(x_i') + \varepsilon_i), \\ V_i^{\mathrm{TD3}} &= \min_{k \in \{1,2\}} Q_{\bar{\phi}_k}(x_i', \widetilde{a}_i'), \end{aligned} \tag{5}$$

$$\begin{aligned} a_i' &\sim \pi_\theta(\cdot \mid x_i'), \\ V_i^{\mathrm{SAC}} &= \min_{k \in \{1,2\}} Q_{\bar{\phi}_k}(x_i', a_i') - \alpha \log \pi_\theta(a_i' \mid x_i'). \end{aligned} \tag{6}$$

SABER does not change any quantity in Eqs. (2)–(6). For critic k, the replay-action Bellman residual remains

$$\delta_{ik} = Q_{\phi_k}(x_i, a_i^r) - y_i^{\mathsf{A}}. \tag{7}$$

Under ordinary uniformly sampled HER, every squared residual has explicit coefficient one.

1.3 Current-Policy Local Support

For each relabeled sample, SABER forms the full discrete observation–goal–horizon context

$$c_i = C(o_i, \widetilde{g}_i, h_i), \tag{8}$$

where h_i is the remaining horizon. A lower-dimensional retrieval key $C^{\mathrm{ret}}(o_i, h_i)$ is used only to query the replay index. Let $\mathbf{J}_i^{\mathrm{ret}}$ be the resulting ordered list of currently valid, unique replay slots. Invalid entries and duplicate slot identifiers are removed in maintained index order, retaining the first valid occurrence. The bounded scan is

$$S_i = \mathrm{Tail}_{N_{\mathrm{scan}}}\left(\mathbf{J}_i^{\mathrm{ret}}\right), \qquad |S_i| \le N_{\mathrm{scan}}. \tag{9}$$

The tail refers to deterministic maintained index order, not guaranteed temporal recency after circular-buffer wraparound. The coarse key is only an acceleration device. Support eligibility is determined by exact component-wise equality of the full context:

$$I_i = \{j \in S_i : C(o_j, \widetilde{g}_i, h_j) = C(o_i, \widetilde{g}_i, h_i)\}. \tag{10}$$

The query's relabeled goal $\widetilde{g}_i$ is used on both sides; a candidate's originally stored desired goal is not used for matching. The sampled transition is not removed from consideration and may therefore occur in I_i.

For action dimension d, let ℓ_d and u_d be its environment bounds. SABER first maps the action to

$$z_d(a) = \mathrm{clip}\left(\frac{a_d - \ell_d}{u_d - \ell_d}, 0, 1\right) \tag{11}$$

and assigns the discrete component

$$b_{K,d}(a) = \min\{K - 1, \lfloor K z_d(a) \rfloor\}. \tag{12}$$

The multidimensional bin $b_K(a)$ is the tuple of these components; the implementation stores its equivalent base-K integer encoding. The local set of occupied replay-action bins is

$$\mathcal{U}_i = \left\{b_K(a_j^r) : j \in I_i\right\}, \qquad n_i = |I_i|. \tag{13}$$

The binary support indicator is

$$\psi_i = \mathbf{1}\{n_i \ge N_{\mathrm{local}}\}\, \mathbf{1}\{b_K(a_i^\pi) \in \mathcal{U}_i\}. \tag{14}$$

If retrieval returns no candidates, exact matching yields too few candidates, or the policy bin is unoccupied, then $\psi_i = 0$. Main SABER uses

$$K = 5, \qquad N_{\mathrm{scan}} = 512, \qquad N_{\mathrm{local}} = 4.$$

DDPG and TD3 use their deterministic actor output as a_i^π. SAC uses the deterministic squashed actor mean for this query only; its target and actor updates continue to use stochastic policy samples. No gradient passes through a_i^π, action binning, retrieval, or ψ_i.

Task family	Full context components	Coarse retrieval key
FetchPickAndPlace	Phase, hand–object distance, object–goal distance, object height, gripper gap, horizon	Hand–object distance, object height, gripper gap, horizon
DrawerOpen/Close	Phase, hand–handle distance, handle–goal distance, handle y, gripper state, horizon	Hand–handle distance, handle y, gripper state, horizon

Table 1: Task-specific support contexts. Candidate retrieval uses the coarse key, after which exact component-wise equality on the full discretized context defines the local replay set.

Component	FetchPickAndPlace thresholds	Drawer thresholds
Phase	approach, grasp, lift, place	approach, contact, opening/closing, open/closed
Hand distance	.03, .05, .08, .12, .20, .40	.03, .05, .08, .12, .20, .40
Goal distance	.05, .10, .20, .40, .70	.03, .05, .08, .12, .20, .40, .70
Task coordinate	object z: .43, .48, .55, .65, .75	handle y: $-.05, 0, .05, .10, .15, .25$
Gripper	gap: .02, .05, .08	state: $-.50, 0, .50$
Remaining horizon	$10, 25, 50, 100$	$10, 25, 50, 100, 200, 500$

Table 2: Discretization boundaries used to construct support contexts. Values are applied component-wise in environment units. The main method normalizes each action dimension to its action bounds and uses $K = 5$ bins per dimension, $N_{\mathrm{scan}} = 512$, $N_{\mathrm{local}} = 4$, and $\lambda_S = 1$.

For the Drawer phase definitions, let d_h be hand–handle distance and d_g be handle–goal distance. The terminal phase is open or closed when $d_g \le 0.05$; opening or closing requires $d_h \le 0.08$ and $0.05 < d_g \le 0.12$; contact requires $d_h \le 0.08$ and $d_g > 0.12$; remaining samples are approach. Phases use the relabeled training goal.

1.4 Mean-Normalized Critic-Only Objective

SABER assigns the raw coefficient

$$w_i^{\mathrm{raw}} = 1 + \lambda_S \psi_i \tag{15}$$

and normalizes it by the minibatch mean:

$$\bar{w}_B = \frac{1}{|B|} \sum_{j \in B} w_j^{\mathrm{raw}}, \qquad \Omega_i = \frac{w_i^{\mathrm{raw}}}{\max(\epsilon_w, \bar{w}_B)}. \tag{16}$$

We use $\lambda_S = 1$ and $\epsilon_w = 10^{-8}$. The safeguard is inactive because $w_i^{\mathrm{raw}} \geq 1$. If $p_B = |B|^{-1} \sum_i \psi_i$, then

$$\Omega_i = \begin{cases} \dfrac{1 + \lambda_S}{1 + \lambda_S p_B}, & \psi_i = 1, \\ \dfrac{1}{1 + \lambda_S p_B}, & \psi_i = 0. \end{cases} \tag{17}$$

Hence every sample remains in the update with a positive coefficient, the minibatch mean coefficient is one, and support-positive samples receive relative weight $1 + \lambda_S$ compared with support-negative samples. If all indicators in a minibatch are equal, then every normalized coefficient is exactly one.

The critic objective is

$$\mathcal{L}_Q^{\mathrm{SABER}} = \frac{1}{|B|} \sum_{i \in B} \Omega_i \sum_{k=1}^{n_Q} \delta_{ik}^2. \tag{18}$$

For TD3 and SAC, the same Ω_i multiplies both critic residuals. The main variant is critic-only:

$$\mathcal{L}_\pi^{\mathrm{SABER}} = \mathcal{L}_\pi^{\mathsf{A}}. \tag{19}$$

Mean-one normalization preserves the average explicit coefficient, but does not imply an unchanged numerical loss or gradient norm because support may correlate with residual magnitude.

Weight-range invariant. For the Main choice $\lambda_S = 1$, the support-negative and support-positive coefficients in any mixed minibatch with $0 < p_B < 1$ satisfy

$$\tfrac{1}{2} < \Omega_i^- < 1 < \Omega_i^+ < 2, \qquad \sum_{i \in B} \Omega_i = |B|.$$

Thus SABER neither drops a sample nor assigns an unbounded coefficient. It reduces exactly to the uniformly weighted HER critic objective when all indicators are zero or all are one.

1.5 End-to-End Update

Algorithm 1 states one complete learning update. The replay index is maintained as transitions enter or overwrite replay slots. Candidate retrieval supplies only a bounded list; exact full-context matching determines support eligibility.

Algorithm 1: One SABER update

Require: replay $\mathcal{D}$ and index $\mathcal{R}$; actor π_θ; critics $\{Q_{\phi_k}\}_{k=1}^{n_Q}$; target networks
Require: context maps C, C^{ret}; $K, N_{\mathrm{scan}}, N_{\mathrm{local}}, \lambda_S$

$B \leftarrow \mathrm{UniformSample}(\mathcal{D})$
Apply Eq. (2) where selected and eligible; recompute rewards.
for each relabeled sample $i \in B$ **do**
  Construct x_i, x_i', bootstrap mask χ_i, and detached y_i^{A} using Eqs. (3)–(6).
  $a_i^\pi \leftarrow \mathrm{sg}[\mathrm{QueryAction}_{\mathsf{A}}(x_i)]$
  $q_i^{\mathrm{ret}} \leftarrow C^{\mathrm{ret}}(o_i, h_i)$
  $\mathbf{J}_i^{\mathrm{ret}} \leftarrow \mathcal{R}[q_i^{\mathrm{ret}}]$
  Remove invalid and duplicate slots from $\mathbf{J}_i^{\mathrm{ret}}$.
  $S_i \leftarrow \mathrm{Tail}_{N_{\mathrm{scan}}}(\mathbf{J}_i^{\mathrm{ret}})$
  $I_i \leftarrow \{j \in S_i : C(o_j, \widetilde{g}_i, h_j) = C(o_i, \widetilde{g}_i, h_i)\}$
  $\mathcal{U}_i \leftarrow \{b_K(a_j^r) : j \in I_i\}$
  $\psi_i \leftarrow \mathbf{1}\{|I_i| \geq N_{\mathrm{local}}\}\mathbf{1}\{b_K(a_i^\pi) \in \mathcal{U}_i\}$
  $w_i^{\mathrm{raw}} \leftarrow 1 + \lambda_S \psi_i$
end for
$\Omega_i \leftarrow w_i^{\mathrm{raw}} / \max(\epsilon_w, |B|^{-1} \sum_{j \in B} w_j^{\mathrm{raw}})$ for every i.
Compute replay-action residuals using Eq. (7).
Update all critics with Eq. (18).
Apply the backbone's unchanged actor, temperature, delayed-update, and target-network rules.

The algorithm introduces no auxiliary target and never regresses the critic at a_i^π. It does not alter minibatch membership, HER relabeling, reward recomputation, exploration, network architecture, optimizer, or UTD. The indicator is a binary occupancy proxy, not an estimate of density, uncertainty, target accuracy, target variance, or TD-error rank. Finally, SABER cannot mark an action as locally supported unless its action bin is already represented by replay actions in the matched context.

Operational and cost boundary. The support path is a detached, non-neural lookup and introduces no trainable parameter or optimizer state. For minibatch B, at most $|B|N_{\text{scan}}$ slots enter exact full-context matching and action-bin occupancy after retrieval; under the Main setting this is at most $256 \times 512 = 131{,}072$ slot checks per update. This bound applies to the post-retrieval scan, not to all index maintenance or candidate-list processing, whose cost can depend on retrieval-key occupancy. Once Ω_i is formed, the backbone follows its ordinary critic, actor, temperature, delayed-update, and target-network rules.

2 Experimental and Reproducibility Details

2.1 Tasks, Cohorts, and Shared Training Settings

Purpose. This section fixes the experimental scope before any method is compared. Its purpose is to identify which quantities are paired, which quantities are shared, and which task–backbone combinations support a matched SABER claim.

Protocol. Tables 3–5 give the settings shared within each matched cohort. The primary comparison contains seven task–backbone cohorts and uses paired seeds 1–5. Drawer SAC is not a primary cohort because a complete matched HER–SABER pair was not collected.

Task	Horizon	Budget	Backbones	Exact primary grid
FetchPickAndPlace-v4	50	1M	DDPG, TD3, SAC	10k (DDPG/TD3), 50k (SAC)
DrawerOpen-v3	200	2M	DDPG, TD3	10k
DrawerClose-v3	200	2M	DDPG, TD3	10k

Table 3: Primary task and cohort definitions. Every method within a task–backbone cohort uses the same environment-step budget and exact evaluation grid. Primary comparisons use paired seeds 1–5.

Setting	Value or convention
HER	Future-goal convention in Sec. 1.2; nominal $p_{\text{HER}} = 0.8$
Replay	Uniform for HER and SABER; capacity 300k; minibatch size 256
Networks	Actor and critic MLPs with two hidden layers of 256 units
Optimization	Adam, actor and critic learning rates 3×10^{-4}
Discount and targets	$\gamma = 0.98$, Polyak coefficient $\tau = 0.005$
Update schedule	UTD 1; learning begins after 5k steps; random actions for first 10k steps
Termination	Successful relabeled transitions are nonterminal; time limits bootstrap
Evaluation	Deterministic policy, 20 episodes per checkpoint
Statistics	Seed mean and sample SD; paired percentile-bootstrap CI, 10,000 resamples, RNG seed 2027
Curve processing	Exact common checkpoints only; no smoothing, interpolation, extrapolation, or filling

Table 4: Shared training, evaluation, and statistical settings. Matched HER–SABER runs use identical replay and relabeling code.

Backbone	Critic	Exploration or entropy	Backbone-specific settings
DDPG	One critic	Gaussian action noise, SD 0.2	Deterministic actor
TD3	Twin critics	Gaussian action noise, SD 0.2	Target noise 0.2, clip 0.5, policy delay 2
SAC	Twin critics	Automatic entropy tuning	Initial $\alpha = 0.2$, α LR 3×10^{-4}, target entropy -4

Table 5: Backbone-specific settings. SABER preserves each backbone's target construction and actor-update rule.

Tuning boundary. We did not conduct an exhaustive joint hyperparameter search. The Main SABER settings were fixed for the matched comparisons, and no control received variant-specific tuning. The $K \in \{3, 7\}$ runs are sensitivity checks around Main $K = 5$, not a per-task model-selection sweep.

Interpretation boundary. Within each paired cohort, HER and SABER share the task, seed, training budget, replay and relabeling code, backbone, optimizer, exploration process, UTD, and evaluation grid. Their intended algorithmic difference is therefore the per-sample critic-loss coefficient. Absolute AUC values are not compared across tasks because horizons and training budgets differ.

2.2 Comparison Methods

Purpose. The comparison suite separates three operational intervention points after experience collection: hindsight selection, replay-transition selection, and fixed-minibatch critic-loss allocation.

Protocol.

Prioritization controls. HGR prioritizes an episode and then a transition–future-goal pair, multiplies the two normalized importance factors on relabeled samples, and leaves the original-goal coefficient at one. PER samples replay transitions with TD-error priority and annealed importance correction. Both change realized selection, unlike SABER and the TD-error critic-loss weighting control.

Method	Selection changed	Loss coefficient	Correction
HER	None beyond uniform replay and HER relabeling	Uniform	None
SABER	None	Current-policy local-support weight	Unit minibatch mean
HGR	Episode and transition–future-goal pair	HGR importance weight	Two-stage IS
PER	Replay transition	PER importance weight	Annealed IS
TD-error weighting	None	TD-error-derived weight	Unit minibatch mean

Table 6: Operational intervention points. HGR uses priority exponents $\alpha_{\text{episode}} = \alpha_{\text{pair}} = 0.6$, while PER uses $\alpha = 0.6$. Both anneal β linearly from 0.4 to 1.0 over critic-update progress and use priority offset 10^{-6}. TD-error weighting retains uniform replay and is therefore the closest allocation-stage control.

TD-error critic-loss weighting. This control preserves uniform replay and changes only normalized per-sample critic coefficients. DDPG uses its single absolute TD error. SAC uses $\min(|\delta_{i1}|, |\delta_{i2}|)$ as the signal and applies the resulting coefficient to both critic losses. TD3 results are not reported because the stored control used independent target-policy smoothing noise for weight construction and optimized residuals, preventing the intended matched-signal interpretation.

Interpretation boundary. HER is the direct matched baseline. HGR and PER are broader prioritization comparisons because they alter realized selection, whereas TD-error weighting is the closest alternative allocation-signal control where validated. Thus an ordering against HGR or PER does not isolate loss allocation alone, while an ordering against TD-error weighting does not establish superiority over every possible weighting signal.

2.3 Metrics and Statistical Analysis

Purpose. The metrics distinguish full-training performance, late-training performance, final-checkpoint behavior, and threshold-reaching time without smoothing away seed variability.

Protocol. Let $q_j \in [0, 1]$ be deterministic evaluation success at exact shared checkpoint t_j. Normalized success AUC is

$$\text{AUC} = \frac{1}{t_m - t_1} \sum_{j=1}^{m-1} \frac{q_j + q_{j+1}}{2} (t_{j+1} - t_j).$$

Final-quarter AUC applies the same calculation to 750k–1M for FPP and 1.5M–2M for Drawer tasks. Final success is the value at the final checkpoint. Hit90 is the first exact checkpoint with $q_j \geq 0.9$; its median is computed only over seeds that reach the threshold, so reached-seed coverage is always reported.

Checkpoint alignment and unit of inference. For seed s, checkpoint success is the mean of 20 binary outcomes from deterministic-policy evaluation episodes. Before computing a comparison, we restrict every included method to the intersection of its available checkpoint labels; AUC, FQ-AUC, and final success for that comparison use this same grid. The statistical unit is the seed rather than an episode or checkpoint. Thus a paired comparison first forms one difference d_s per shared seed, and the paired bootstrap resamples the vector of seed-indexed differences. Evaluation episodes and checkpoints are never treated as independent replicates. A seed that does not reach success 0.9 contributes to reached-seed coverage but is not assigned an artificial Hit90 time.

Evaluation isolation. Evaluation uses the deterministic policy with exploration disabled and a seed stream separate from training. Evaluation transitions are not inserted into replay and trigger no optimizer or target-network update. Checkpoint labels count training environment interactions only, so evaluation frequency does not change the stated training budget or replay contents.

Cohort separation. Primary claims use the paired five-seed cohorts only. The frozen FPP TD3 ten-seed cohort, indexing sensitivity, and Drawer diagnostic seeds 6–10 are reported as confirmatory, sensitivity, or diagnostic extensions and are not silently pooled into primary confidence intervals. Where a pooled ten-seed diagnostic summary is shown, it is labeled descriptive.

For paired comparisons, each seed contributes one method difference. The reported 95% intervals are percentile intervals from 10,000 paired bootstrap resamples with pseudorandom seed 2027. Sample standard deviations use $n - 1$ in the denominator. No curve is smoothed, interpolated, extrapolated, or forward-filled.

Metric and uncertainty boundary. The trapezoidal AUC weights each interval by elapsed environment steps, so a denser checkpoint schedule does not receive extra mass merely by containing more checkpoints. Normalized AUC and FQ-AUC remain in $[0, 1]$. Method-wise mean and sample SD describe across-seed level and dispersion, whereas paired intervals quantify uncertainty in within-seed differences. An interval that includes zero is treated as inconclusive rather than as evidence of equivalence. Hit90 uses only an observed checkpoint crossing and is never interpolated between evaluations.

AUC boundary handling. The full-span integral begins at the first shared recorded checkpoint t_1 and ends at t_m; we do not prepend a synthetic zero-step success value. FQ-AUC restricts the same checkpoint sequence to the stated task-specific window and normalizes by that window's observed elapsed span. The exact grids contain the window endpoints, so neither metric requires a fractional or interpolated checkpoint. Final success is q_m on the same shared grid. These rules are applied unchanged to every method in a reported comparison.

Interpretation boundary. Where reported, Steps/s is the seed-1–5 mean of completed training steps divided by runtime. Because historical runs were not hardware-normalized across families, it is descriptive rather than a learning-efficiency claim. Control and reference tables use a common metric order; diagnostic tables retain signal-specific columns.

3 Primary Comparison Details

3.1 Drawer Learning Curves

Purpose. Full-span AUC compresses the learning trajectory into one number. The Drawer curves expose when success emerges, whether gains persist late in training, and how much of each aggregate ordering is driven by seed variability.

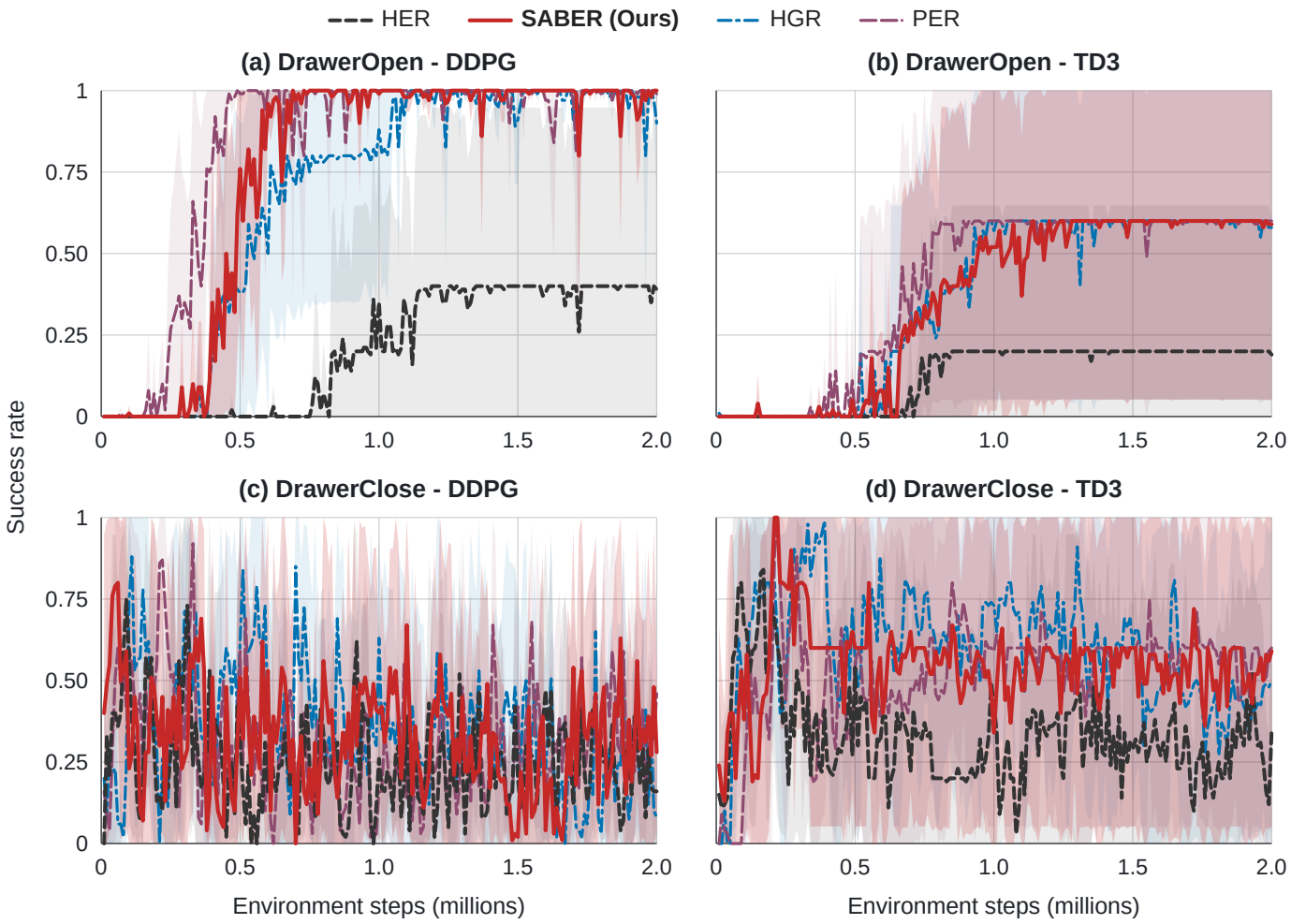


Figure 1: DrawerOpen and DrawerClose learning curves for DDPG and TD3. Lines are seed means and bands are sample standard deviations over paired seeds 1–5 on the exact 10k grid. No smoothing or interpolation is applied.

Results. DrawerOpen DDPG shows the clearest separation from HER, whereas the TD3 cohorts and comparisons with HGR and PER exhibit larger seed dispersion and mixed metric-dependent orderings. On DrawerClose DDPG, HGR leads full-span AUC but not FQ-AUC, illustrating that early and late summaries can disagree.

Interpretation. The trajectories support the matched HER result but do not establish a universal ranking against HGR or PER. They also show why full-span AUC, FQ-AUC, and final success are retained separately: early learning, late occupancy, and a single final checkpoint need not order methods identically.

3.2 Time to Threshold

Purpose. Hit90 tests whether a method reaches a practically high success level and how early it first does so, complementing metrics that average over checkpoints.

Results. Across the three FPP backbones, SABER reaches Hit90 in all 15 runs, compared with 11/15 for HER, 5/15 for HGR, and 10/15 for PER. On DrawerOpen DDPG, SABER reaches in 5/5 seeds at a reached-seed median of 500k, whereas HER reaches in 2/5 at 960k. On DrawerClose, every primary method reaches Hit90 in all seeds for both backbones, despite substantial differences in later performance.

Interpretation. Hit90 is a threshold-crossing summary rather than a stability measure. Its conditional median is meaningful only together with reached-seed coverage, and early threshold crossing cannot substitute for FQ-AUC or final success.

Task	Backbone	Method	Median Hit90	Reached seeds	Steps/s
FetchPickAndPlace	DDPG	SABER	540k	5/5	41.7
		HER	600k	5/5	65.0
		HGR	930k	2/5	16.9
		PER	640k	4/5	58.0
FetchPickAndPlace	TD3	SABER	840k	5/5	41.2
		HER	980k	3/5	65.3
		HGR	630k	1/5	12.9
		PER	1.00M	1/5	50.8
FetchPickAndPlace	SAC	SABER	750k	5/5	29.1
		HER	800k	3/5	38.9
		HGR	800k	2/5	8.4
		PER	550k	5/5	34.8
DrawerOpen	DDPG	SABER	500k	5/5	23.3
		HER	960k	2/5	41.1
		HGR	530k	5/5	13.2
		PER	330k	5/5	34.0
DrawerOpen	TD3	SABER	780k	3/5	21.6
		HER	740k	1/5	41.5
		HGR	760k	3/5	11.6
		PER	720k	3/5	34.6
DrawerClose	DDPG	SABER	20k	5/5	24.0
		HER	70k	5/5	41.2
		HGR	100k	5/5	13.2
		PER	60k	5/5	33.9
DrawerClose	TD3	SABER	80k	5/5	22.5
		HER	70k	5/5	41.7
		HGR	80k	5/5	11.8
		PER	120k	5/5	34.4

Table 7: Hit90 for the primary comparisons. Hit90 is the first exact-grid checkpoint at which deterministic evaluation success reaches at least 0.9. Medians use only reached seeds, so reached-seed coverage is reported separately. Steps/s is the descriptive seed mean defined in Section 1.3.

3.3 FPP TD3 Confirmatory and HER-Indexing Sensitivity Results

Purpose. The ten-seed extension tests whether the primary five-seed FPP TD3 direction persists under a larger frozen cohort. The indexing sensitivity asks whether that direction is specific to excluding $m(s_{t+1})$ from future goals.

Results. In the ten-seed cohort, SABER exceeds HER in 8/10 pairs, with mean AUC difference $+0.0467$, paired-bootstrap 95% CI $[0.0085, 0.0783]$, and exact sign-flip $p = 0.0527$. Under inclusive-future indexing, the five-seed mean difference is $+0.0674$, with 3/5 positive pairs, 95% CI $[-0.0068, 0.1416]$, and sign-flip $p = 0.25$, compared with $+0.0708$ under the primary convention.

Interpretation. The extension narrows the estimated TD3 gain while preserving positive paired bootstrap interval evidence. The inclusive variant preserves the observed mean direction, but it is a five-seed sensitivity rather than an equivalence test between indexing conventions.

Seed	HER AUC	SABER AUC	Difference
1	0.3465	0.4301	$+0.0836$
2	0.3727	0.4568	$+0.0841$
3	0.3942	0.4960	$+0.1018$
4	0.4109	0.4066	-0.0043
5	0.3338	0.4225	$+0.0886$
6	0.4301	0.5207	$+0.0907$
7	0.3995	0.4614	$+0.0619$
8	0.3068	0.3543	$+0.0475$
9	0.4106	0.4114	$+0.0008$
10	0.4210	0.3331	-0.0879
Mean	0.3826	0.4293	$+0.0467$

Table 8: Frozen FPP TD3 confirmatory extension over ten paired seeds. AUC uses the exact 10k–1M grid without smoothing, interpolation, or filling.

Seed	HER AUC	SABER AUC	Difference
1	0.3005	0.4576	$+0.1571$
2	0.2674	0.2669	-0.0005
3	0.3467	0.5179	$+0.1712$
4	0.3487	0.4003	$+0.0515$
5	0.3669	0.3247	-0.0422
Mean	0.3261	0.3935	$+0.0674$

Table 9: FPP TD3 inclusive-future sensitivity, also admitting $m(s_{t+1})$. AUC uses the exact 10k–1M grid. FQ-AUC: $+0.1854$ (95% CI $[-0.0212, 0.3920]$); final: $+0.2200$ (95% CI $[-0.2000, 0.5200]$).

4 Additional Controls and Sensitivities

4.1 Allocation Signal

Purpose. This control asks whether SABER benefits merely from assigning unequal critic coefficients, or whether current-policy local support provides a different allocation signal from TD error. Both methods retain uniform replay and change only normalized critic-loss coefficients.

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s	ΔAUC
FPP–DDPG	SABER (Main)	0.609 ± 0.038	0.961 ± 0.014	0.960 ± 0.042	5/5 (0.540M)	41.7	$--$
	TD-error weighting	0.568 ± 0.069	0.910 ± 0.119	0.930 ± 0.104	5/5 (0.630M)	61.1	$+0.041\ [-0.018, 0.123]$
DrawerOpen–DDPG	SABER (Main)	0.754 ± 0.044	0.987 ± 0.018	1.000 ± 0.000	5/5 (0.500M)	23.3	$--$
	TD-error weighting	0.715 ± 0.090	0.988 ± 0.011	0.800 ± 0.447	5/5 (0.610M)	39.0	$+0.038\ [-0.015, 0.091]$
DrawerClose–DDPG	SABER (Main)	0.324 ± 0.082	0.277 ± 0.100	0.280 ± 0.251	5/5 (0.020M)	24.0	$--$
	TD-error weighting	0.313 ± 0.098	0.351 ± 0.218	0.400 ± 0.166	5/5 (0.130M)	38.9	$+0.011\ [-0.079, 0.074]$
FPP–SAC	SABER (Main)	0.583 ± 0.030	0.986 ± 0.011	0.990 ± 0.022	5/5 (0.750M)	29.1	$--$
	TD-error weighting	0.544 ± 0.068	0.901 ± 0.167	0.990 ± 0.022	5/5 (0.800M)	38.2	$+0.040\ [-0.016, 0.095]$

Table 10: Validated TD-error critic-loss weighting controls over paired seeds 1–5. Replay remains uniform, and only critic-loss coefficients change. Values are mean ± sample SD. The final column reports SABER-minus-control paired AUC differences with 95% bootstrap CIs; positive values favor SABER. For SAC, the priority signal is $\min(|\delta_{i1}|, |\delta_{i2}|)$, and the resulting normalized coefficient is applied to both critic losses. Steps/s is the descriptive seed mean defined in Section 1.3.

Results. SABER has higher observed mean AUC in all four validated cohorts, but every paired 95% interval includes zero. Late metrics are mixed, including higher FQ-AUC for TD-error weighting on DrawerClose.

Interpretation. The common mean direction distinguishes SABER from this specific TD-error allocation control, but the five-seed intervals do not establish a conclusive per-cohort advantage. TD3 is omitted because its stored control does not support the intended matched-signal comparison.

4.2 Action-Bin Resolution

Purpose. This sensitivity tests whether the Main action discretization $K = 5$ is robust to coarser and finer bins and records how bin resolution changes the frequency of support activation.

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s	Support+	ΔAUC
FPP–TD3	$K = 3$	0.353 ± 0.048	0.603 ± 0.121	0.750 ± 0.265	2/5 (0.945M)	33.9	0.396 ± 0.012	$+0.089\ [0.070, 0.109]$
	$K = 5$ (Main)	0.442 ± 0.035	0.804 ± 0.103	0.970 ± 0.045	5/5 (0.840M)	41.2	0.219 ± 0.009	$--$
	$K = 7$	0.361 ± 0.047	0.616 ± 0.151	0.690 ± 0.263	2/5 (0.890M)	36.1	0.148 ± 0.010	$+0.081\ [0.067, 0.093]$

Table 11: FPP TD3 action-bin sensitivity over paired seeds 1–5. Values are mean ± sample SD. Support+ is the support-positive rate. The final column reports Main-minus-variant paired AUC differences with 10,000-resample percentile-bootstrap 95% CIs; positive values favor Main $K = 5$.

Results. Main $K = 5$ has higher mean AUC than both $K = 3$ and $K = 7$, with both Main-minus-variant 95% intervals excluding zero. The support-positive rate decreases as K increases.

Interpretation. The evaluated $K = 5$ setting is favored over both adjacent resolutions in this cohort. Because K simultaneously changes action-bin geometry and activation density, the experiment does not identify either factor alone or establish $K = 5$ as universally optimal.

4.3 Actor and Critic Weight Scope

Purpose. This ablation tests whether the support coefficient should modify the critic, the actor, or both. All variants retain the support query, support hyperparameters, and minibatch normalization. Critic-Only weights Bellman losses; Actor-Only weights per-sample actor terms; Actor+Critic weights both.

Results. Critic-Only has the highest observed mean AUC in each of the five completed Main-versus-scope comparisons. Main-minus-variant differences range from $+0.017$ to $+0.141$, but every 95% interval includes zero. DrawerOpen Actor-Only also has substantial seed dispersion (0.612 ± 0.343). FPP Actor+Critic remains pending and is not inferred.

Interpretation. The available results are consistent with using critic-only weighting as the conservative main design, while providing no decisive evidence that actor weighting is intrinsically harmful. Each scope changes the subsequent training trajectory, so realized support masks need not remain identical.

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s	ΔAUC (Main–variant)
FPP–DDPG	Critic-Only (Main)	0.609 ± 0.038	0.961 ± 0.014	0.960 ± 0.042	5/5 (0.540M)	41.7	$--$
	Actor-Only	0.589 ± 0.032	0.952 ± 0.016	0.960 ± 0.022	5/5 (0.600M)	15.8	$+0.020[-0.024, 0.063]$
	Actor+Critic			*pending*			
DrawerOpen–DDPG	Critic-Only (Main)	0.754 ± 0.044	0.987 ± 0.018	1.000 ± 0.000	5/5 (0.500M)	23.3	$--$
	Actor-Only	0.612 ± 0.343	0.760 ± 0.433	0.800 ± 0.447	4/5 (0.470M)	18.6	$+0.141[-0.051, 0.478]$
	Actor+Critic	0.728 ± 0.084	0.992 ± 0.008	0.990 ± 0.022	5/5 (0.540M)	18.4	$+0.026[-0.048, 0.091]$
DrawerClose–DDPG	Critic-Only (Main)	0.324 ± 0.082	0.277 ± 0.100	0.280 ± 0.251	5/5 (0.020M)	24.0	$--$
	Actor-Only	0.307 ± 0.039	0.299 ± 0.061	0.280 ± 0.299	5/5 (0.040M)	24.0	$+0.017[-0.026, 0.078]$
	Actor+Critic	0.298 ± 0.071	0.241 ± 0.119	0.440 ± 0.370	5/5 (0.050M)	23.7	$+0.026[-0.043, 0.091]$

Table 12: DDPG actor-scope ablations over paired seeds 1–5. Values are mean ± sample standard deviation. Hit90 reports the number of reaching seeds and the reached-seed median step. Steps/s is the descriptive seed mean defined in Section 1.3. The final column gives Main-minus-variant paired AUC differences with 95% bootstrap confidence intervals. Pending cells are left unreported until a complete five-seed cohort is available.

4.4 Secondary Replay and Update-Budget References

Purpose. These references place SABER outside the strictly matched UTD-1 comparison: the no-HER comparison separates uniform experience replay (ER), uniform replay with AER's actor/critic learning-rate schedule (Annealing ER), and Attentive Experience Replay (AER). AER ranks an enlarged uniform candidate set by cosine similarity to the current online state and trains on the 256 most similar transitions. The three arms therefore test attentive selection beyond its shared schedule and whether replay selection alone can solve sparse FPP. High-replay-ratio HER and REDQ-HER instead test larger update budgets and critic ensembles. REDQ-HER TQB clips the sampled target-critic backup to the sparse-reward range $[-50, 0]$ before forming the Bellman target.

Results. Without HER, AER/Annealing ER/ER obtain mean AUCs of $0.044/0.129/0.148$ with DDPG, $0.039/0.039/0.039$ with TD3, and $0.040/0.041/0.064$ with SAC. AER therefore has no higher observed mean AUC than either replay control in these three backbones. On FPP SAC, high-replay-ratio HER and REDQ-HER reach AUC 0.768 and 0.867, respectively, above SABER's 0.583, but use UTD 4 and REDQ also uses ten critics. On DrawerClose, REDQ-HER reaches Hit90 in all seeds at 20k but records AUC/FQ-AUC $0.188/0.098$, compared with $0.324/0.277$ for DDPG SABER.

Interpretation. In these matched transfers, state-similarity replay neither substitutes for hindsight relabeling nor improves mean AUC over schedule-matched uniform replay. This is not a general replication or rejection of AER; it isolates transferred AER components under the present FPP backbones. Increased update budget and ensemble size can be stronger on FPP SAC but are not uniformly stronger across tasks. Because HER use, UTD, algorithm, and architecture differ, these tables are scaling and context references rather than one-factor tests of SABER.

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s
FPP–DDPG	ER	0.148 ± 0.039	0.354 ± 0.124	0.460 ± 0.188	0/5	68.4
	Annealing ER	0.129 ± 0.111	0.259 ± 0.251	0.340 ± 0.363	1/5 (0.960M)	68.0
	AER	0.044 ± 0.002	0.045 ± 0.004	0.000 ± 0.000	0/5	67.0
FPP–TD3	ER	0.039 ± 0.002	0.043 ± 0.002	0.000 ± 0.000	0/5	21.0
	Annealing ER	0.039 ± 0.002	0.043 ± 0.002	0.000 ± 0.000	0/5	19.6
	AER	0.039 ± 0.002	0.043 ± 0.002	0.000 ± 0.000	0/5	26.2
FPP–SAC	ER	0.064 ± 0.042	0.114 ± 0.135	0.130 ± 0.199	0/5	39.4
	Annealing ER	0.041 ± 0.002	0.047 ± 0.008	0.000 ± 0.000	0/5	39.5
	AER	0.040 ± 0.003	0.044 ± 0.002	0.000 ± 0.000	0/5	39.3

Table 13: FPP no-HER replay references over seeds 1–5. Values are mean ± sample SD on the exact 10k–1M grid. These runs omit hindsight relabeling and are context references rather than matched post-HER mechanism controls. Steps/s is the descriptive seed mean defined in Section 1.3.

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s
FPP–SAC	HER UTD 1; 2 critics	0.471 ± 0.079	0.735 ± 0.247	0.790 ± 0.288	3/5 (0.800M)	38.9
	SABER UTD 1; 2 critics	0.583 ± 0.030	0.986 ± 0.011	0.990 ± 0.022	5/5 (0.750M)	29.1
	HER high-RR UTD 4; 2 critics	0.768 ± 0.097	0.963 ± 0.021	0.960 ± 0.042	5/5 (0.450M)	5.9
	REDQ-HER TQB					
FPP–REDQ	UTD 4; 10 critics/2 target	0.867 ± 0.052	0.993 ± 0.008	1.000 ± 0.000	5/5 (0.300M)	3.1

Table 14: FPP SAC and REDQ scaling references on the exact shared 50k–1M grid. Values are mean ± sample SD over seeds 1–5. High-replay-ratio HER and REDQ-HER change the update budget, and REDQ-HER also changes critic architecture; they are not matched post-HER loss-allocation controls. Steps/s is the descriptive seed mean defined in Section 1.3.

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s
DrawerClose–DDPG	HER UTD 1; 1 critic	0.251 ± 0.030	0.250 ± 0.093	0.160 ± 0.182	5/5 (0.070M)	41.2
	SABER UTD 1; 1 critic	0.324 ± 0.082	0.277 ± 0.100	0.280 ± 0.251	5/5 (0.020M)	24.0
	REDQ-HER TQB					
DrawerClose–REDQ	UTD 4; 10 critics/2 target	0.188 ± 0.041	0.098 ± 0.022	0.100 ± 0.117	5/5 (0.020M)	5.8

Table 15: DrawerClose update-budget and architecture reference over seeds 1–5. Values are mean ± sample standard deviation on the exact 10k–2M grid. FQ-AUC covers 1.5M–2M; Final is final-checkpoint success; Hit90 gives reaching seeds and their median first-hit step. REDQ-HER uses four updates per environment step and a 10-critic ensemble with two target critics sampled per backup, so it is not a resource-matched SABER comparison. Steps/s is the descriptive seed mean defined in Section 1.3.

5 Support-Signal Diagnostic Details

5.1 Instrumentation and Aggregation

Purpose. The diagnostics test whether SABER's support rule activates in practice, whether it is redundant with TD-error ranking, and whether the resulting weights alter immediate critic fitting in the intended direction.

Protocol. Diagnostic logging introduces no gradient term and does not change replay sampling, objectives, or parameter-update order. FPP logs one full minibatch every 10k environment steps for DDPG, TD3, and SAC seeds 1–5: 100 batches per run, 500 per backbone, and 1,500 total. Drawer diagnostics use a 200-step cadence from 5k through 2M, producing 9,976 persisted snapshots per seed. Snapshot quantities are aggregated within seed before cohort statistics are computed.

For each persisted minibatch, the diagnostic stores the pre-update absolute Bellman residual, support membership, and the residual after one critic update. The fixed-target residual is

$$\Delta|\delta_i| = \left|Q_\phi^{\mathrm{pre}}(x_i, a_i^r) - y_i^{\mathrm{pre}}\right| - \left|Q_\phi^{\mathrm{post}}(x_i, a_i^r) - y_i^{\mathrm{pre}}\right|.$$

Thus the target and minibatch are held fixed while the critic parameters change. Live-target quantities, where available, are retained only for audit and are not mixed with the fixed-target statistic.

Interpretation boundary. These measurements audit activation and one-update optimization behavior on persisted minibatches. They do not by themselves identify the independent cause of a long-horizon return difference.

5.2 Cohort-Level Results

Purpose. The cohort summary jointly measures activation frequency, association with pre-update error, and support-conditioned residual reduction under a common fixed-target definition.

Results. Primary-cohort support-positive rates range from 0.197 to 0.417, so the indicator is neither nearly absent nor nearly constant. Primary Pearson correlations are modest (0.084–0.143), while exact top-decile enrichment ranges from $1.379\times$ to $1.828\times$.

Cohort	Seeds	Support+ rate	Pearson r	Top-10% enrich.	Pre-update $\lvert\delta\rvert$ $(+/-)$	One-step $\Delta\lvert\delta\rvert$ $(+/-)$
FPP–DDPG	1–5	0.367 ± 0.011	0.116	$1.409\times$	$1.42\times$	$2.39\times$
FPP–TD3	1–5	0.219 ± 0.009	0.128	$1.670\times$	$1.62\times$	$3.30\times$
FPP–SAC	1–5	0.197 ± 0.013	0.143	$1.828\times$	$1.74\times$	$3.94\times$
DrawerOpen–DDPG	1–5	0.417 ± 0.021	0.084	$1.379\times$	$1.16\times$	$1.79\times$
DrawerOpen–DDPG	6–10 extension	0.475 ± 0.094	0.095	$1.440\times$	$1.13\times$	$1.91\times$
DrawerOpen–DDPG	1–10 pooled	0.446 ± 0.071	0.090	$1.409\times$	$1.15\times$	$1.83\times$
DrawerClose–DDPG	1–5	0.298 ± 0.018	0.104	$1.534\times$	$1.42\times$	$2.94\times$
DrawerClose–DDPG	6–10 extension	0.299 ± 0.007	0.118	$1.602\times$	$1.48\times$	$4.23\times$
DrawerClose–DDPG	1–10 pooled	0.298 ± 0.013	0.111	$1.568\times$	$1.45\times$	$3.48\times$

Table 16: Support-signal diagnostics. Statistics are first aggregated within seed, after which seeds are independent units. Support rates are mean $\pm$ sample SD. The last two columns divide support-positive by support-negative cohort means. One-step $\Delta|\delta|$ evaluates the same minibatch after one critic step while retaining the pre-update Bellman target. Seeds 1–5 are the primary diagnostic cohorts, seeds 6–10 are separate extensions, and pooled ten-seed rows are descriptive.

The support-positive/support-negative one-step residual-reduction ratio ranges from $1.79\times$ to $3.94\times$ in the primary cohorts. Both Drawer extensions preserve enrichment above one and a larger reduction for support-positive samples.

Interpretation. Support-positive samples are enriched among high-error samples, but the weak linear correlations show that support does not reduce to a TD-error ranking. The one-step statistic verifies that the normalized coefficients redistribute immediate critic fitting as intended; it is not a standalone explanation of long-term learning performance.

5.3 Phase-Conditioned Allocation

Purpose. The phase analysis asks where support activation occurs during Drawer progression and whether it merely follows the most frequent phase.

Cohort	Phase	Sample fraction	Support+ rate	Seeds
DrawerOpen–DDPG	Approach	0.174 ± 0.018	0.745 ± 0.022	1–5
	Contact	0.049 ± 0.003	0.698 ± 0.056	
	Opening	0.046 ± 0.005	0.636 ± 0.053	
	Open	0.730 ± 0.018	0.306 ± 0.027	
DrawerOpen–DDPG	Approach	0.195 ± 0.011	0.754 ± 0.030	6–10 ext.
	Contact	0.027 ± 0.024	0.515 ± 0.181	
	Opening	0.023 ± 0.023	0.440 ± 0.263	
	Open	0.755 ± 0.039	0.387 ± 0.127	
DrawerClose–DDPG	Approach	0.218 ± 0.002	0.549 ± 0.032	1–5
	Contact	0.016 ± 0.003	0.710 ± 0.026	
	Closing	0.019 ± 0.004	0.425 ± 0.034	
	Closed	0.747 ± 0.004	0.212 ± 0.017	
DrawerClose–DDPG	Approach	0.222 ± 0.003	0.569 ± 0.004	6–10 ext.
	Contact	0.018 ± 0.003	0.727 ± 0.016	
	Closing	0.017 ± 0.002	0.438 ± 0.028	
	Closed	0.743 ± 0.007	0.204 ± 0.010	

Table 17: Drawer phase-conditioned allocation. Phases are defined from the relabeled training goal. Values are seed means $\pm$ sample SD; support is sample-weighted within each seed. Seeds 6–10 are separate extensions. Pooled ten-seed and seed-level values are included in the artifacts.

Results. In both Drawer tasks and in both primary and extension cohorts, approach and contact have higher support-positive rates than the dominant terminal phase, even though the terminal phase accounts for roughly three quarters of logged samples.

Interpretation. Activation therefore does not simply track phase frequency. The task-specific phase labels localize the signal but do not constitute an independent causal test of long-horizon performance.

6 Runtime and Resource Measurements

6.1 Execution-Environment Audit

Purpose. The audit records the execution boundary needed to assess paired fairness and to distinguish software-controlled comparisons from hardware-specific throughput observations.

Results. All four hosts used the verified common experiment-library stack: Python 3.12.3, PyTorch 2.12.1+cu130, CUDA 13.0, cuDNN 9.20.0, Gymnasium 1.3.0, Gymnasium-Robotics 1.4.2, MuJoCo 3.3.0, Meta-World 3.1.1, NumPy 2.3.5, and SciPy 1.18.0. For each paired HER–SABER cohort, both methods and all paired seeds were executed on one physical host under this stack; no paired cohort was split across machines. Sanitized provenance is recorded in `execution_host_mapping.csv`.

Host	System	Compute hardware	OS, kernel, and driver
A	MSI GB10 system	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10	Ubuntu 24.04.4; 6.17.0-1014-nvidia; driver 580.142
B	GIGABYTE AI TOP ATOM	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10	Ubuntu 24.04.4; 6.17.0-1026-nvidia; driver 580.159.03
C	GIGABYTE AI TOP ATOM	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10	Ubuntu 24.04.4; 6.17.0-1026-nvidia; driver 580.159.03
D	GB10 system	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10	Ubuntu 24.04.3; 6.11.0-1016-nvidia; driver 580.95.05

Table 18: Hardware inventory for the four physically distinct execution hosts. All systems use CUDA-addressable unified memory rather than dedicated VRAM. Hosts B and C are separate machines with the same model and system configuration. Hostnames, serial numbers, and other identifying metadata are omitted.

Interpretation. The paired performance differences are not created by moving HER and SABER across machines within a cohort. Hardware still differs across cohorts, so historical throughput is descriptive and is not used for cross-host ranking.

6.2 Dedicated Runtime Profile

Purpose. The dedicated profile quantifies the current implementation overhead under a sequential same-hardware comparison and identifies which pipeline component dominates that overhead. It runs HER and SABER for FPP TD3 seed 1 over the 20k–100k interval.

Profile item	Recorded value
Execution protocol	Sequential FPP TD3 HER and SABER, seed 1, 20k–100k
GPU	NVIDIA GB10, compute capability 12.1, 121.7 GiB CUDA-addressable memory
CPU/platform	20 logical CPUs, AArch64 Linux 6.17.0-1026-nvidia, glibc 2.39
Software	Python 3.12.3, PyTorch 2.12.1+cu130, CUDA 13.0, cuDNN 9.20.0
Comparison boundary	The dedicated profile is not extrapolated to historical primary runs
Memory boundary	Peak process and accelerator memory were not measured

Table 19: Recorded infrastructure for the dedicated same-hardware runtime profile. CUDA-addressable memory is reported because the GB10 uses unified memory. These identifiers characterize the dedicated runtime profile; historical steps/s values in other tables are descriptive cohort measurements.

Component	HER ms/step	SABER ms/step	Difference	SABER share
Interaction	1.000	1.071	$+0.070$	4.4%
Replay + HER	1.925	2.651	$+0.726$	11.0%
Batch preparation	0.096	0.108	$+0.012$	0.4%
Support and weighting	0.000	15.028	$+15.028$	62.3%
Critic update	2.945	2.832	-0.113	11.7%
Actor + target update	1.384	1.292	-0.091	5.4%
Evaluation/logging amortized	0.080	0.085	$+0.005$	0.4%
Other loop overhead	0.633	1.066	$+0.434$	4.4%
Total training loop	8.063	24.133	$+16.070$	100.0%

Table 20: Hardware-specific component times for the FPP TD3 seed-1 implementation profile over 20k–100k steps. Other loop overhead is the difference between total loop time and the explicitly instrumented components.

Results. HER and SABER achieve 124.0 and 41.4 environment steps/s, corresponding to 8.063 and 24.133 ms per training step. Support and weighting account for 15.028 ms/step, or 62.3% of SABER's measured loop time, and explain most of the $+16.070$ ms/step difference.

Interpretation. The profile attributes the dominant measured cost to the current support retrieval implementation, motivating faster retrieval. It is one hardware-specific, early-training profile rather than an asymptotic or end-to-end learning-efficiency result. Historical steps/s values were not collected under this sequential protocol, and peak memory was not measured; neither quantity is inferred.

7 Artifact and Provenance Index

Purpose. This index connects each reported experiment to a validation status and repository-relative artifact location, so completed evidence is distinguishable from references, extensions, and pending work.

Content	Artifact key	Status
Primary seed metrics and curves	`primary_seed_results`	Complete
Action-bin sensitivity	`action_bin_sensitivity`	Complete
DDPG actor-scope analysis	`actor_scope_ddpg`	FPP A+C pending
Derived Technical Supplement summaries	`technical_supplement`	FPP A+C pending
FPP diagnostics	`fpp_diagnostic`	Complete
DrawerClose diagnostics	`drawerclose_diagnostic`	Complete
DrawerOpen fixed-target diagnostics	`draweropen_fixed_target`	Complete
FPP TD3 runtime profile	`fpp_td3_runtime`	Complete
Execution-environment audit	`execution_environment_audit`	Complete
FPP TD3 ten-seed confirmation	`fpp_td3_n10`	Complete
FPP TD3 inclusive-future sensitivity	`fpp_td3_inclusive`	Complete
FPP SAC update-budget references	`fpp_sac_scaling`	Complete
DrawerClose REDQ scaling reference	`drawerclose_redq`	Complete

Table 21: Artifact index. Keys resolve to repository-relative package paths in `artifact_index.csv`. Canonical artifacts include manifests that bind source files by repository-relative path and SHA256.

Seed mapping. For paired seed s, the environment, PyTorch, NumPy, and action-noise streams use s; the support-weight stream uses $20000 + s$, and evaluation uses $10000 + s$. Training and evaluation resets advance by deterministic episode- and checkpoint-indexed offsets. All methods in a paired cohort use this mapping.

Interpretation boundary. Generated analysis directories contain compact CSV summaries, a repository-relative source manifest, and a self-excluding artifact manifest. Canonical experiment artifacts retain configuration, environment snapshot, source provenance, and validation records where available. Raw result files are referenced rather than duplicated in the derived summaries. An indexed artifact establishes traceability to the reported result; it does not convert a secondary reference into a matched causal comparison.

A Primary Seed-Level Results

Purpose. The seed-level appendix exposes every observation underlying the primary aggregate table, allowing readers to inspect paired directions and between-seed dispersion directly.

Results. SABER has higher observed mean AUC than matched HER in all seven primary cohorts, and four paired 95% intervals exclude zero. The individual rows show that this aggregate statement coexists with negative or near-zero seed pairs and especially large dispersion on the Drawer TD3 cohorts. AUC and final-quarter AUC use each cohort's exact common grid.

Interpretation. These rows are the observations used in the primary analysis, not an additional independent experiment. They make the direction and uncertainty of each paired cohort auditable without changing the seed as the unit of statistical analysis.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.640152	0.956000	0.900	0.510M
	2	0.537374	0.934000	0.950	0.600M
	3	0.578788	0.964000	0.950	0.610M
	4	0.325253	0.731000	0.900	0.920M
	5	0.612374	0.967000	1.000	0.570M
SABER	1	0.656061	0.939000	0.950	0.480M
	2	0.615152	0.972000	0.950	0.540M
	3	0.564899	0.959000	1.000	0.630M
	4	0.631818	0.976000	1.000	0.480M
	5	0.577525	0.960000	0.900	0.610M
HGR	1	0.441667	0.765000	1.000	0.870M
	2	0.076515	0.065000	0.000	–
	3	0.374495	0.482000	0.500	–
	4	0.417677	0.587000	0.700	0.990M
	5	0.380556	0.506000	0.500	–
PER	1	0.484091	0.919000	1.000	0.770M
	2	0.520455	0.953000	1.000	0.730M
	3	0.604293	0.946000	0.900	0.550M
	4	0.643434	0.964000	1.000	0.460M
	5	0.049242	0.046000	0.000	–

Table 22: Seed-level primary results for FPP with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.346465	0.523000	0.550	–
	2	0.372727	0.635000	0.950	1.000M
	3	0.394192	0.662000	1.000	0.980M
	4	0.410859	0.686000	0.950	0.970M
	5	0.333838	0.530000	0.750	–
SABER	1	0.430051	0.846000	1.000	0.840M
	2	0.456818	0.856000	1.000	0.860M
	3	0.495960	0.914000	0.950	0.630M
	4	0.406566	0.652000	0.900	0.630M
	5	0.422475	0.752000	1.000	0.910M
HGR	1	0.345202	0.515000	0.500	–
	2	0.396212	0.593000	0.800	–
	3	0.407576	0.581000	0.850	–
	4	0.326515	0.489000	0.450	0.630M
	5	0.361111	0.493000	0.450	–
PER	1	0.345455	0.515000	0.550	–
	2	0.408586	0.653000	0.950	1.000M
	3	0.394192	0.628000	0.800	–
	4	0.369192	0.526000	0.500	–
	5	0.327020	0.488000	0.500	–

Table 23: Seed-level primary results for FPP with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.559211	0.985000	1.000	0.800M
	2	0.469737	0.760000	1.000	0.950M
	3	0.406579	0.500000	0.500	–
	4	0.540789	0.965000	1.000	0.800M
	5	0.378947	0.465000	0.450	–
SABER	1	0.636842	0.995000	0.950	0.650M
	2	0.567105	0.980000	1.000	0.800M
	3	0.573684	0.980000	1.000	0.750M
	4	0.575000	1.000000	1.000	0.750M
	5	0.564474	0.975000	1.000	0.800M
HGR	1	0.451316	0.545000	0.800	–
	2	0.596053	0.980000	1.000	0.700M
	3	0.506579	0.800000	1.000	0.900M
	4	0.450000	0.505000	0.500	–
	5	0.414474	0.490000	0.500	–
PER	1	0.755263	0.970000	1.000	0.450M
	2	0.635526	0.985000	1.000	0.550M
	3	0.706579	0.990000	1.000	0.550M
	4	0.690789	1.000000	1.000	0.550M
	5	0.660526	0.985000	0.950	0.600M

Table 24: Seed-level primary results for FPP with SAC. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.460302	0.970000	1.000	1.090M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.000000	0.000000	0.000	–
	5	0.576256	0.993500	0.950	0.830M
SABER	1	0.757286	1.000000	1.000	0.510M
	2	0.786935	0.977000	1.000	0.420M
	3	0.705025	1.000000	1.000	0.580M
	4	0.713065	0.960000	1.000	0.500M
	5	0.806030	1.000000	1.000	0.400M
HGR	1	0.799749	0.995000	0.500	0.410M
	2	0.704523	0.990000	1.000	0.530M
	3	0.692211	0.993000	1.000	0.610M
	4	0.794975	0.998000	1.000	0.430M
	5	0.449497	0.934500	1.000	1.060M
PER	1	0.859799	1.000000	1.000	0.330M
	2	0.877387	0.998000	1.000	0.270M
	3	0.814070	0.969000	1.000	0.330M
	4	0.809171	0.951500	0.950	0.390M
	5	0.765075	0.990000	1.000	0.410M

Table 25: Seed-level primary results for DrawerOpen with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.632286	0.999500	0.950	0.740M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.000251	0.000000	0.000	–
	5	0.000000	0.000000	0.000	–
SABER	1	0.486558	0.996500	0.950	1.070M
	2	0.000000	0.000000	0.000	–
	3	0.000503	0.000000	0.000	–
	4	0.689950	0.997000	1.000	0.560M
	5	0.632161	0.997000	1.000	0.780M
HGR	1	0.000000	0.000000	0.000	–
	2	0.692462	0.992000	1.000	0.520M
	3	0.531910	0.997000	0.900	0.930M
	4	0.000000	0.000000	0.000	–
	5	0.628266	0.996000	1.000	0.760M
PER	1	0.632412	0.986000	1.000	0.780M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.755528	0.999000	1.000	0.520M
	5	0.644221	0.994000	1.000	0.720M

Table 26: Seed-level primary results for DrawerOpen with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.281658	0.337500	0.300	0.090M
	2	0.268844	0.315500	0.100	0.150M
	3	0.203518	0.193000	0.400	0.040M
	4	0.249497	0.289000	0.000	0.070M
	5	0.251508	0.116000	0.000	0.020M
SABER	1	0.287814	0.165500	0.650	0.010M
	2	0.337437	0.417000	0.200	0.060M
	3	0.284045	0.306500	0.150	0.050M
	4	0.461307	0.300500	0.000	0.020M
	5	0.251508	0.197500	0.400	0.010M
HGR	1	0.672613	0.218000	0.000	0.100M
	2	0.323869	0.236500	0.000	0.090M
	3	0.237563	0.154500	0.150	0.120M
	4	0.238568	0.219500	0.050	0.100M
	5	0.273367	0.299000	0.200	0.010M
PER	1	0.288568	0.272000	0.550	0.100M
	2	0.244975	0.273500	0.000	0.180M
	3	0.297990	0.392000	1.000	0.040M
	4	0.331281	0.356500	0.350	0.050M
	5	0.376884	0.404000	0.400	0.060M

Table 27: Seed-level primary results for DrawerClose with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.450628	0.455000	0.850	0.080M
	2	0.058291	0.002000	0.000	0.120M
	3	0.886181	0.847000	0.900	0.050M
	4	0.097236	0.017000	0.000	0.060M
	5	0.151382	0.098000	0.000	0.070M
SABER	1	0.080653	0.020000	0.000	0.080M
	2	0.908291	0.978000	1.000	0.090M
	3	0.739950	0.761000	1.000	0.200M
	4	0.908920	0.909500	0.950	0.010M
	5	0.106030	0.000000	0.000	0.050M
HGR	1	0.923618	0.886000	0.400	0.080M
	2	0.686307	0.808500	0.950	0.060M
	3	0.782161	0.428000	1.000	0.130M
	4	0.477010	0.115000	0.150	0.060M
	5	0.138945	0.063000	0.000	0.260M
PER	1	0.111055	0.050000	0.000	0.220M
	2	0.845477	0.987000	1.000	0.190M
	3	0.824874	0.891000	1.000	0.120M
	4	0.112060	0.002000	0.000	0.120M
	5	0.725377	0.983000	1.000	0.100M

Table 28: Seed-level primary results for DrawerClose with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.